

# **Lecture 39 — Least squares, data fitting, and positive definiteness**

Wed Dec 2

Last updated: October 8, 2026.

Reading: Boyd 12.2–12.4 and 13.1, class notes on symmetric and positive-definite matrices, and Lachniet Ch. 2 (least-squares / data-fitting exercises). Related objectives: [Lectures](#). [Schedule](#).

Bring paper and a laptop if you have one. Warm-up, fill-in, a small real dataset on the board/software, then **Your turn**. Full least-squares software practice is [Lab 6](#) on **Friday** (same day as [Quiz 10](#) — Quiz 10 is Week 13 eigenvalues, **not** today's fitting).

## Learning objectives

By the end of class, students should be able to:

1. Compute a least-squares solution with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check the residual's orthogonality to the columns of  $A$ .
5. Evaluate whether a fitted model is useful rather than merely reporting coefficients.
6. Recognize that  $A^T A$  is symmetric positive semidefinite, explain  $x^T A^T A x = \|Ax\|^2$ , and recognize that  $A^T A$  is positive definite when  $A$  has linearly independent columns.

## Warm-up

Monday's setup: minimize  $\|Ax - b\|$ . At a minimizer  $\hat{x}$ , the residual  $r = b - A\hat{x}$  is orthogonal to  $C(A)$ , which is equivalent to the **normal equations**

$$A^T A \hat{x} = A^T b.$$

- (a) If  $A$  is  $m \times n$  with  $m > n$ , what size is  $A^T A$ ? Why is that nicer than solving  $Ax = b$  directly?
- (b) True or false:  $A^T A$  is always symmetric.

## Fill in

**Normal equations / software.** Solving  $A^T A \hat{x} = A^T b$  (when  $A$  has independent columns) gives the unique least-squares  $\hat{x}$ . In practice we often call a solver instead of forming  $A^T A$  by hand:

| Task                    | Octave                     | Python                                                   |
|-------------------------|----------------------------|----------------------------------------------------------|
| Least-squares $\hat{x}$ | <code>xhat = A \ b</code>  | <code>xhat = np.linalg.lstsq(A, b, rcond=None)[0]</code> |
| Fitted values           | <code>bhat = A*xhat</code> | <code>bhat = A @ xhat</code>                             |
| Residual                | <code>r = b - bhat</code>  | <code>r = b - bhat</code>                                |

| Task | Octave                             | Python                                              |
|------|------------------------------------|-----------------------------------------------------|
| RMS  | <code>rms = norm(r)/sqrt(m)</code> | <code>rms =<br/>np.linalg.norm(r)/np.sqrt(m)</code> |

**Orthogonality check.** At a least-squares solution,  $A^T r \approx$  \_\_\_\_\_ (numerical noise ok).

**RMS.** Root-mean-square residual  $\text{RMS} = \|r\|/\sqrt{m}$  reports a typical per-equation error in the same units as  $b$ .

**Useful vs coefficients.** Always ask: do fitted values track the data? Is RMS small *in context*? Blindly quoting  $\hat{x}$  is not enough.

**Positive semidefinite / definite (Boyd / class notes).** A symmetric matrix  $P$  is **positive semidefinite (PSD)** if  $x^T P x \geq 0$  for all  $x$ ; **positive definite (PD)** if  $x^T P x > 0$  for all  $x \neq 0$ .

For any  $A$ :

$$x^T (A^T A) x = \|Ax\|^2 \geq 0,$$

so  $A^T A$  is always \_\_\_\_\_. It is **positive definite** precisely when  $Ax = 0 \Rightarrow x = 0$ , i.e. when  $A$  has \_\_\_\_\_ columns — then the normal equations have a unique solution.

## Worked in class

**W1.** Tiny fit by hand / software. Four campus shuttle wait times (minutes) at hours  $t = 1, 2, 3, 4$  after 8am:

$$b = (6, 7, 10, 11).$$

Model a line  $b \approx x_1 + x_2 t$  with

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}.$$

- Form  $A^T A$  and  $A^T b$  (or use `A\b` / `lstsq`).
- Report  $\hat{x} = (x_1, x_2)$  and the fitted values  $\hat{b} = A\hat{x}$ .
- Compute  $r$  and RMS. In one sentence: is a linear trend reasonable here?

**Expected (for checking):**  $\hat{x} = (4, 1.8)$  exactly from the normal equations;  $\hat{b} = (5.8, 7.6, 9.4, 11.2)$ ;  $\text{RMS} = \sqrt{0.2} \approx 0.45$  minutes.

**W2.** Orthogonality. With your  $\hat{x}$  from W1, compute  $A^\top r$ . Should it be near  $(0, 0)$ ?

**W3.** Plot story. On axes “hour  $t$ ” vs “wait (min)”, sketch data points and the fitted line. Where is the residual for  $t = 1$  visible?

**W4.** PSD identity. Expand  $x^\top A^\top A x$  for  $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$  and a general  $x = (x_1, x_2)$ . Confirm it equals  $\|Ax\|^2$  and is  $\geq 0$ .

**W5.** PD vs PSD. Explain why  $A^\top A$  fails to be positive definite if the two columns of  $A$  are equal. What goes wrong for uniqueness of  $\hat{x}$ ?

## Your turn

**Y1.** Using the W1 data, suppose software returns  $\hat{x} = (4, 1.8)$ . Compute  $\hat{b}_1 = x_1 + x_2 \cdot 1$  and the first residual  $r_1 = b_1 - \hat{b}_1$ .

**Y2.** True or false:  $A^T A$  is always positive definite for every  $A$ .

**Y3.** Why check  $A^T r \approx 0$  after a least-squares solve?

**Y4.** A fit has  $\text{RMS} = 0.02$  °C for outdoor temperature data that typically varies by 10 °C across the day. Useful model, or not? One sentence.

## Answers (Your turn)

Y1.  $\hat{b}_1 = 4 + 1.8 = 5.8$ ;  $r_1 = 6 - 5.8 = 0.2$ .

Y2. **False.**  $A^\top A$  is always PSD, but PD only when  $A$  has linearly independent columns (equivalently  $\ker A = \{0\}$ ).

Y3. It is the normal-equation / geometric check that the residual is orthogonal to  $C(A)$  — evidence that  $\hat{x}$  really is a least-squares critical point (up to roundoff).

Y4. **Useful** in context:  $\text{RMS} \ll$  the natural scale of the data, so typical pointwise error is tiny compared with day-scale variation.